

Homework 5 - Model Misspecification

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Task 1: Misspecifying the Deterministic Part

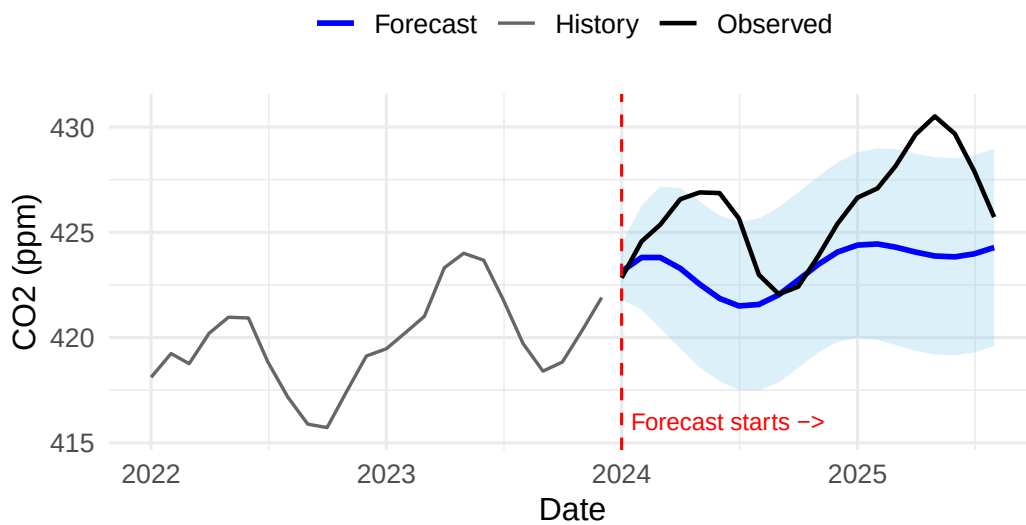
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CO2 Misspec. Task 1 – (Linear μ only) – Holdout 2024 (

RMSE = 3.41 ppm | MAE = 2.76 ppm



Question: What happens if the deterministic part is misspecified?

A misspecification of the deterministic part of a model will lead to improper predictions. In particular, we will miss out on important *deterministic* harmonic behavior caused by seasonal effects (like CO2 being absorbed by plants during the Global North's summer-time). It is not immediately obvious that this deterministic harmonic trend is missing, as we still observe harmonic-like behavior in our predictions. As it turns out, those “wiggles” are introduced by the ARIMA! It is capturing some of the seasonal variance that was uncaptured by the linear deterministic trend. This effect helps to keep predictions moving forward in time to assume a reasonable-appearing shape, but with a poorly-fitting deterministic trend and a “confused” ARIMA model attempting to resolve deterministic *and* stochastic variation, we end up with quite consistent underprediction. Analyzing residuals would show that they do not have mean 0.

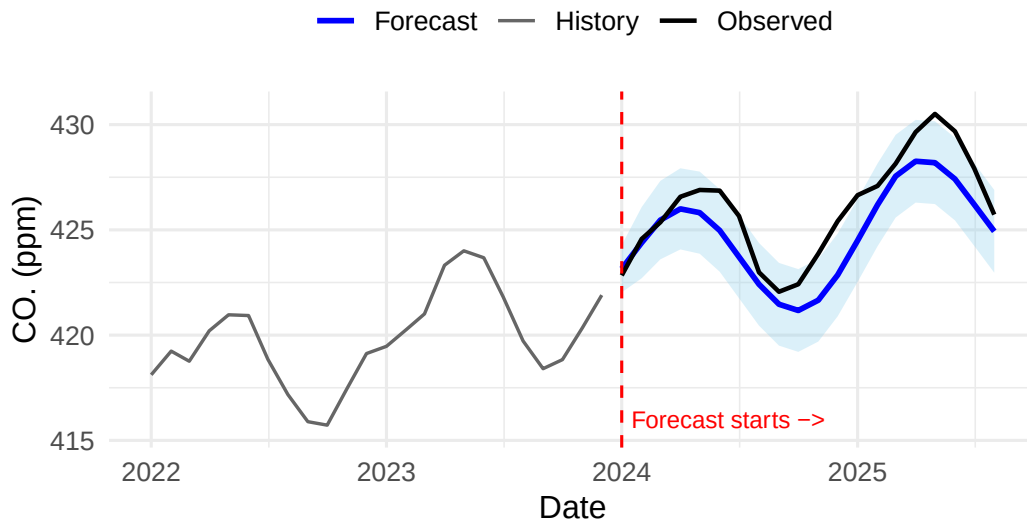
Task 2: Misspecifying the Stochastic Process

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CO2 – Misspec. Task 2 (ARMA(1,1) eps.) – Holdout 2024
RMSE = 1.49 ppm | MAE = 1.27 ppm



Question: What happens if the stochastic process is misspecified?

A misspecification of the stochastic process in this example leads to fairly consistent under-prediction of future CO2. In general, for a time series model, we expect to obtain biased predictions for inflexible specification of the stochastic model. Since our ARIMA model has arguments (1,0,1), we actually have an ARMA model with only first-order autoregression and first-order moving average terms. No differencing is applied. Differencing is important for this example, because we know that our mean CO2 has been reliably increasing and therefore, the mean is nonstationary. While the predictions look fairly acceptable for this short time frame, we would expect our omission of differencing to have large, undesirable predictive bias over the long term. We did not test our assumptions for a reasonable amount of lags to include for AR and MA, so there may be relevant information encoded in other lag variables, another potential misspecification error. Our deterministic component “saves” the predictions here to make them appear reasonable, but the structure of the model is poor. In the next example, we’ll see what happens if the deterministic component does not come to the rescue.

Task 3: Misspecifying Deterministic, Stochastic Parts

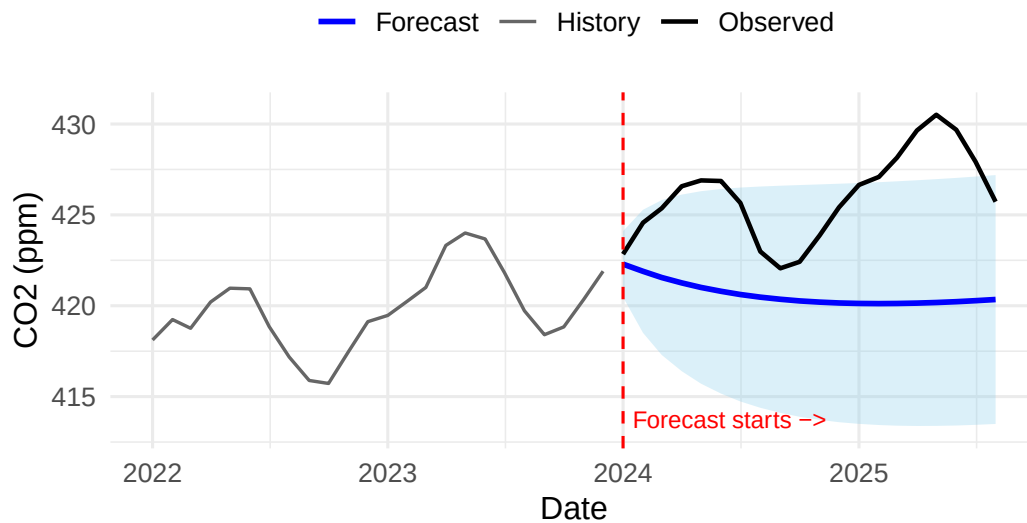
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CO2 – Misspec. Task 3 (Linear mu + ARMA(1,1)) – 2024

RMSE = 6.05 ppm | MAE = 5.42 ppm



Question: What happens if both parts are misspecified?

Misspecification of *both* the deterministic and stochastic parts will lead to predictive behavior that is very inconsistent with reality, and therefore, poor predictions.

In this example, we are missing quadratic and harmonic deterministic elements, and differencing and deeper lag variables for ARIMA. The result is an incredibly poor prediction. As we can see, in the absence of a well-defined stochastic model or a well-defined deterministic model, we lose the shape of the trend entirely. The interpretability of the model also fades, as it becomes less clear just what the ARMA model is trying to capture.

Misspecification of both deterministic and stochastic parts can be very unforgiving. As we see here, misspecifying both parts leads to significantly worse predictions than misspecifying either part. Our error metrics flag this model as the worst by far.

Credit to Dr. Salvana on much of the code (from Lec3.R) used to generate this assignment.